

The Efficiency of Ensemble Machine Learning Models on Network Intrusion Detection using KDDCup 99 Dataset

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Abstract— With the advent of data communication the increased usage of the technologies results in network intrusions and associated attacks. Consequently, the data violation rates are increased abundantly and that sacrifices Confidentiality, Integrity and Availability. This article focused on the network Intrusion Detection System (IDS) that detects various attacks and types. Machine learning (ML) has the potential to spot known-experience and Zero-day attacks. Consequently, the article has considered ML and ensemble models for the various attack classification. The major contributions of the current article are 3-fold. Initially, to understand the relevance and sufficiency of the dataset through exploratory data analysis. Second, the comprehensive understanding of the various attacks, its nature, various types and classifications and finally, the empirical analysis of the dataset through the potential of various ML models. The article utilized various discriminative models for the execution and all of the models have shown better accuracy. The tree-based ensemble model, Random Forest has outperformed the rest of the models with higher accuracy in the training and testing samples of 99.997% and 99.969% respectively.

Keywords—Network Intrusion Detection System (NIDS), Naïve Bayes (NB), Logistic Regression (LR), Decision Tree (DT), Random Forest (RF), Gradient Boosting (GB), Support Vector Machine (SVM), Artificial Neural Network(ANN).

I. INTRODUCTION

The recent developments in the area of internet and communication technologies have led to a huge increase in the network size and the associated data as well as the malicious activities on it. As a result, many assaults, including zero-day attacks, are created, posing significant difficulties for network security to effectively identify these intrusions. The major concern in security is the confirming of the CIA triads – Confidentiality, Integrity and Availability. An IDS facilitates the network to prevent possible malicious activities or intrusions by monitoring the flow of network to confirm CIA triads. An IDS scans the network for the presence of any potential malicious activities or data breaches and also monitors the network packets. IDSs are mainly bifurcated into two types – Deployment Method-based IDS and Detection Method-based IDS. The Network-based IDS (NIDS), Host-based IDS (HIDS), Protocol-based IDS (PIDS), Application Protocol-based IDS (APIDS) and Hybrid IDS (HIDS) are belonging to Deployment Method-based IDS.

All devices connected to the network are examined by NIDSs, which also place a special emphasis on the traffic that travels across the entire subnet and check its compatibility with a database of known or tested threats. The network's independent hosts run Host IDS, which examines incoming and outgoing data packets from the device and alerts the network administrator if any suspicious or malicious activity is detected. The protocol used by a user/device and the server is under the control and interpretation of protocol-based IDSs. By regularly checking the HTTPS protocol and approving the other HTTP-related protocols, PIDS secures the web server. Application Protocol-based IDS is a group of servers that monitors and analyses traffic in order to find intrusions in application-specific protocols. In order to create a comprehensive picture of the network, hybrid IDS combines data from two or more IDS classes, combining network information with host agent or system data. The Hybrid IDS's ability to integrate several IDS strategies is what makes it effective.

There are various methods to detect intrusions such as signature-based methods and anomaly-based methods. These methods are belonging to the Detection Method-based IDS. The signature-based methods detect the patterns or signatures that exist in the IDS. Mostly these patterns are in the form of a stream of bytes in 0's and 1's. Consequently, these methods are poor in the detection of zero-day attacks. Anomaly-based methods, in contrast to signature-based methods, use ML models and other techniques to identify unknown or zero-day threats.

II. LITERATURE REVIEW

Review of the Literature section constitutes various recent research articles that contributed to IDS and also discusses the associated technologies used in the article. The research article [1] focused on the IDS on the popular KDD Cup dataset. The study has two stages to detect and classify the attacks. First Stage utilized the KNN algorithm to identify the vulnerabilities and the second stage used the RF, J48, Adaptive Boosting and NB models. Among these models, the ensemble ML model RF provided a respectable accuracy of 99.97%. The study [2] was conducted using the DNN approaches on the IDS using the six different datasets - KDD Cup99, NSL-KDD, UNSW-NB15, WSN-DS, CICIDS 2017 and Kyoto dataset by Kyoto University. Research Paper [3] incorporates a novel IDS approach that combines the DT and rules-based approaches. In order to extract the input features from the data set and classify the network traffic as Attack/Benign, the study made use of the

capabilities of the REP Tree and JRip algorithm. By examining the CICIDS2017 dataset, the Forest PA classifier was also applied to the data set in addition to the results of the first two classifiers.

The study [4] used the ML algorithms SVM (SVM) and NB on the NSL-KDD Dataset. Among the algorithms, SVM has shown respectable results with 93.95. The article [5] utilized the Feed Forward - Convolutional Neural Network algorithm for the Intrusion Network Detection on NSL-KDD Dataset. Paper [6] employed ML classifiers including SVM (SVM), K-Nearest Neighbor (KNN), LR, NB, Multi-layer Perceptron (MLP), RF, Extra Tree classifier (ETC) and DT for the classification of NSL-KDD dataset to detect as normal or intrusive. Article [7] presented the comprehensive analysis on KDD99 and UNSW-NB15 datasets using three met heuristic algorithms - a rough-set theory (RST), a back-propagation NN (BPNN), and a discrete variant of the cuttlefish algorithm (D-CFA). The study [8] examines the classification on the KDDCup99 dataset using Bayes Net, J48, RF, and Random Tree using the Weka experiment tool. The ensemble ML model RF provided better accuracy.

The research article [9] proposed a novel Multi-tree algorithm and an ensemble adaptive voting algorithm and also utilized some base classifiers - DT, RF, kNN and DNN. The empirical results of these algorithms provided better accuracy for a multi-tree and an ensemble adaptive voting algorithms with 84.2% and 85.2% respectively. The study [10] examines the real-time network IDS using the KDDCup99 using the DL model AEAlexJNET and provided a respectable accurateness of 94.32%. The research also proposed a novel method for the IDS dimension reduction method with a self-encoder. The study used the Flume tool as the agent for the real-time log collection. The paper [11] presents a DL approach for NIDS on NSL-KDD Dataset and the study provides better accuracy in Signature-Based Intrusion Detection with reduced False Positive and Negative rates. The article [12] proposed a novel SwiftIDS, which analyzes the traffic and performance of massive network data. The paper presents two approaches - a light GB machine (LightGBM) and a parallel ID mechanism. LightGBM is used to find efficient ID performance and to analyze the traffic of the network data. For the analysis the study used three bench-mark datasets - KDD99, NSL-KDD and CICIDS2017 and SwiftIDS has been shown an improved accuracy than the existing models.

The research [13] proposed Feed-Forward Deep Neural Network (FFDNN) for the wireless IDS system. A Wrapper Based Feature Extraction Unit (WFEU) has been utilized for the feature extraction with the Extra Trees. The effectiveness and efficiency of the algorithm evaluated on the NSW-NB15 and the AWID IDS datasets using the methods - RF, SVM, NB, DT and kNN. Article [14] utilized the spark MLlib ML Library for anomaly detection and DL methods such as Convolutional Auto-encoder for efficient ID on the heterogeneous CSE-CIC-IDS2018 dataset. The paper [15] examines the processing of massive network traffic data by employing the Apache Spark big data tool, in order to process enormous amounts of data. The article proposes a hybrid methodology in ML and DL. The analysis has employed the stacked auto-encoder for latent feature extraction. The article proceeds through the IDS classification methods such as SVM, RF, DT and NB for the

efficient classification and detection of the network data using UNB ISCX 2012 dataset.

The paper [16] proposed an efficient Transformer-based ID System (RTIDS) for the dimensionality reduction and feature extraction for the imbalanced datasets and the study also examines the sequential information between the features by employing the positional embedding. To understand low-dimensional features from high-dimensional input, the resilience of the stacked encoder-decoder NN is used. The network kinds are identified using the self-attention method. The rigorous real-time IDS tests were conducted on the two benchmark datasets, CICIDS2017 and CIC-DDoS2019, which had respective F1-Scores of 99.17% and 98.48%. The study [17] provided a new IDS framework for feature selection using the integrated ensemble approaches in combination with C4.5 and RF by Penalizing Attributes. The paper also compares the standard ML and DL algorithms such as SVM, RNN, FNN, and LSTM. The research employed the two heuristic algorithms for dimensionality reduction - Correlation-based Feature Selection (CFS) and Bat Algorithm (BA) on the popular benchmark datasets for IDS - NSL-KDD, AWID, and CIC-IDS2017. The empirical results had been showing the robustness of the algorithms.

III. METHODOLOGY

The methodology section constitutes two subsections – Dataset Description and Methodology for IDS. The dataset description incorporated the elaborated analysis of the features of the dataset, statistical analysis, parameter features and attack category. Then the various ML, ensemble, DL and transformer-based models were used for the IDS on NSL-KDD.

A. Dataset Description

Knowledge Discovery and Data Mining (KDD) refer to a family of datasets, which incorporates DARPA, KDDCUP 99, and NSL_KDD. The KDDCup99 is the valuable dataset for the IDS and dataset created by tcpdump facts of the ID contest DARPA generated by the MIT Lincoln laboratory by employing around 1K UNIX machines. The DARPA incorporates 41 attributes and one responsible variable that represents the attack types. The 41 attributes are divided into three groups: the fundamental properties of individual TCP connections (9 properties), the content properties within a connection determined by domain knowledge (13 properties), and the traffic properties determined using a 2s time window (19 properties). The dataset contains around #494021 training and #311029 testing instances. The dataset holds some issues in data processing and analysis. First, the training dataset is too large; consequently, it is time-consuming for building models. The solution is considering 10% of the sampling dataset or eliminating the duplicates and outliers from the dataset. The data preprocessing or data wrangling has to be performed for the outlier elimination and noise removal. In the current research, the dataset considered the distribution of training and test data as #145586 and #77291.

There are 23 different attack types under four main classes – Denial of Service (DoS), Probe, Remote to Local (R2L) and User to Root (U2R). DoS attacks is the making the network resources down and degrade the performance. The probe is gathering comprehensive statistics about the system and information about the network settings. R2L is

unauthorized from the remote node and the intruders transmit the packets to the node or server through a communication channel without permission. U2R is the capturing of the authentication of the system illegally. The attacks under the DoS category are back, land, neptune, pod, smurf and teardrop. DoS attacks are brief and need sending a lot of connections to the server in order to analyze it. Large numbers of source and destination bytes are included in "back" or "backtrack," "land" is a connection from the same host, "neptune" sent a lot of requests quickly, "pod" sent a lot of the wrong segments, "smurf" attacks sent a lot of requests quickly, and "teardrop" sent a lot of the incorrect sections through the communication channel. The attacks under the Probe category are ipsweep, nmap, portsweep and satan. Probe attack has a low login successful rate that tends to zero, 'ipsweep' attack has a high percentage of connections to different hosts, 'nmap' gains access to uncontrolled ports on a system, 'portsweep' has the longest duration and source bytes among all attacks and 'satan' attack has the high percentage of connection to different services. The empirical analysis considered 10% of the dataset, and the details of the data set is depicted in Figure 1.

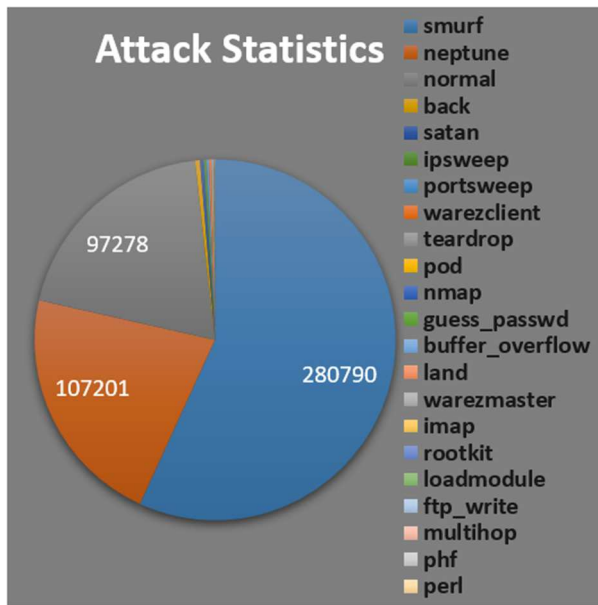


Fig. 1. Statistics of the Attack Types

The attacks under the R2L category are ftp_write, guess_passwd, multihop, imap, phf, spy, warezclient and warezmaster. R2L causes a high number of hot indicators, 'ftp_write' deals with a high number of urgent packets, 'guess_passwd' leads to a high number of failed logins, 'imap' is designed to access plaintext login credentials, 'multihop' is the high number of file creations, 'phf' has a large number of operations on access control files, 'spy' is the spyware attacks are a type of malware and that installed on a system without user authentication, 'warezclient' and 'warezmaster' exploits the vulnerabilities present in "anonymous" File Transfer Protocol on operating systems both Linux and Windows. The attacks under the U2R category are buffer overflow, loadmodule, perl and rootkit. U2R attack takes a large number of root accesses. The 'buffer_overflow' is happening when there is huge data in a buffer than a buffer can handle, 'loadmodule' exploits poor

protection and verification of environment variables, 'perl' provides a loophole to penetrate the system through the websites to the hackers, spammers, and bots and users that unintentionally download dangerous software by opening spam emails are vulnerable to "rootkit" attacks.

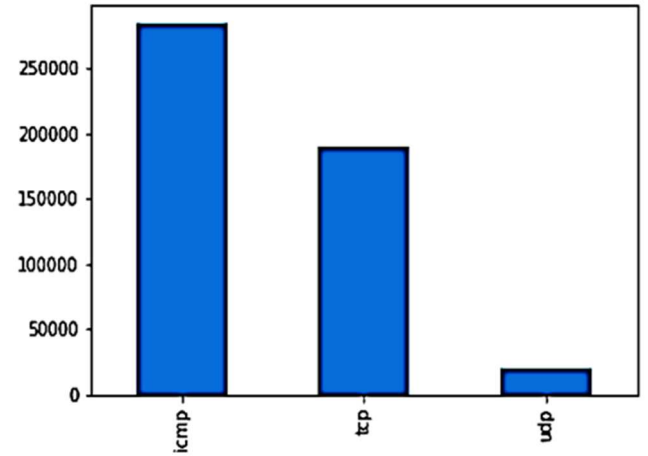


Fig. 2. Features of the Protocol type category

The categorical features of the dataset are categorized into 'protocol_type', 'flag' and 'service'. The categorical features of the 'protocol_type' are icmp, tcp and udp and that is represented in Figure 2. The vulnerabilities are happening through the ports of the TCP and UDP connections. Figure 3 represents the categorical features of the 'flag'. Figure 4 depicts the features of the 'service' category. The correlation between the features has to be detected and the irrelevant features will be removed from the set. Finally, all the features will be mapped to the canonical format through feature mapping.

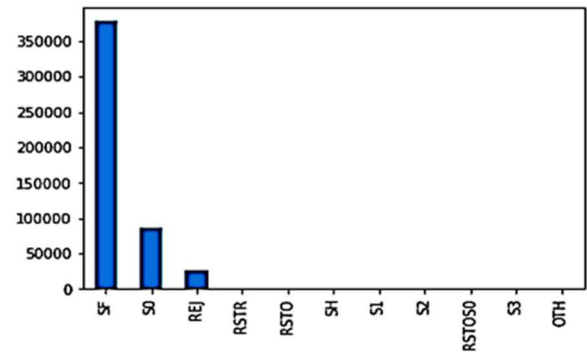


Fig. 3. Features of the Flag category

B. Methodology for IDS

The research incorporated some ML, ensemble and DL models for the development and implementation of the models such as Gaussian NB, LR, DT, RF, GB Classifier, SVM and ANN.

- NB classifiers is a collection of classification algorithms based on Bayes' Theorem and the collection of algorithms adhering to a principle such as every pair of features being classified is independent of each other. It is a probabilistic classifier algorithm based on probabilistic models that constitute strong independence assumptions. The other algorithms used for the empirical analysis are

distributive models, leaving just the NB classifier as a generative model.

- LR Supervised ML model and which calculates the probability of events based on the independent variables in the dataset and the outcome of the algorithm is the categorical dependent variable and the probability of the output variable ranged from 0 to 1. LR can be used for solving both Regression and Classification problems. The model fits the sigmoid or logistic function and predicts 0 or 1.
- A DT, which can be applied to regression or classification models in the form of a tree structure, is favoured for problem-solving in classification. A DT divides a dataset into smaller subsets, and the

output is a tree containing decision nodes and leaf nodes. It is a tree-structured classifier, where internal nodes stand in for a dataset's features, branches for the decision-making process, and each leaf node for the classification result. The dataset's features are used to inform the decisions. The DT algorithm compares the values of the root attribute with the dataset attribute starting at the root node of the tree. The algorithm makes a choice depending on the comparison, then proceeds to the next node by following the branch. The method then compares each node up until the leaf node with a max_depth of 4 with the attribute value.

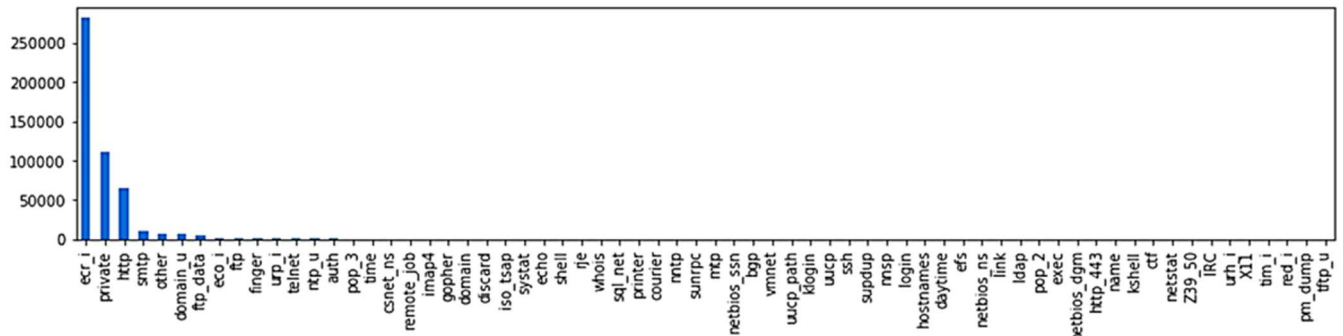


Fig. 4 Features of the 'service' category

- RF classifier is an ensemble model with a subset of DTs and considers the average to accelerate the accuracy by taking the majority votes of predictions from the trees in the forest to get the final output. By the way, the RF classifier leads to improved accuracy and overcomes the problem of overfitting.
- SVM that maps data to a high-dimensional feature space and searches for a hyperplane in an N-dimensional space. A separator between the categories is found, then the data are transformed and a hyperplane could be drawn as a separator.
- A neural network is a group of algorithms and each of that attempts to simulate the network of neurons. ANN facilitates to change of the input consequently the network gives the best result without redesigning the output procedure. In ANN first layer with 30 input dimensions with activation function - "relu", the next layer - "sigmoid" and the final classification layer - "softmax". The optimizer for the model used is "adam".

IV. RESULTS AND DISCUSSION

The section comprise the analysis of the empirical model. The testing set is considered as 33% of the dataset with a random state of 42. As depicted in Figure 5, the generative model NB classifier has shown poor accuracy in both training and testing while comparing the discriminative models. The NB classifier learns both probabilities of attack type and also analyzes the probability between the features and attack type. The reason for the failure of the NB classifier due to the distribution of training and testing data were different, many features are not an independent and

high correlation between the positive and negative features. The rest of the discriminative models has shown respectable accuracy in training and testing. The LR minimized the cost function, utilizing regularization for avoiding overfitting and used the Gradient Descent for the implementation. The ensemble model RF has provided the highest accuracy of 99.997% in training and 99.969 % in testing.

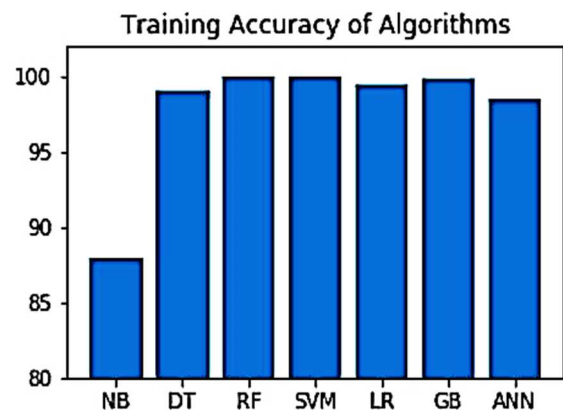


Fig. 5. The training and Testing Accuracy for the models

As represented in Figure 6, GB and ANN are the highest time-consuming algorithms for the training while testing the SVM has the highest (X-axis represents the model names and Y-axis represented the execution time per second).

The accuracy of these models can be increased with eliminate duplicates and data wrangling. SVM is based on the kernel function and that scales with the number of training samples rather badly.

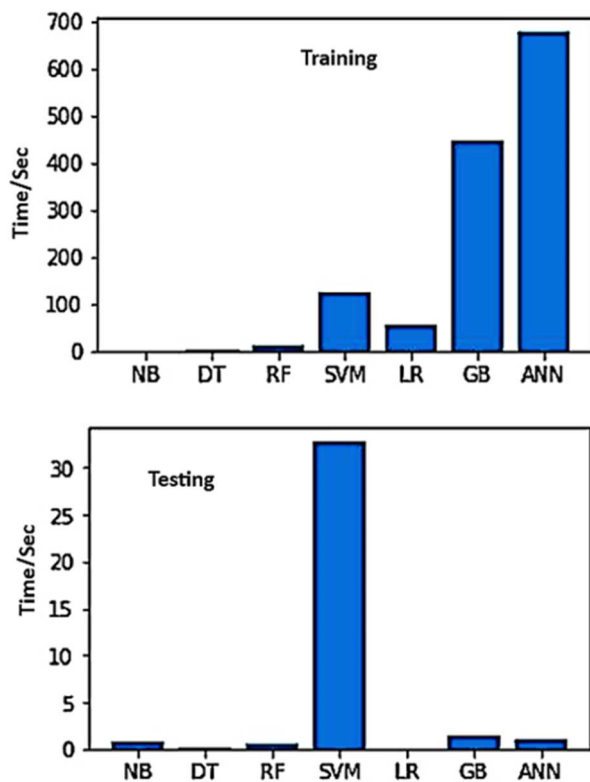


Fig. 6. The training and Testing Time for the models

V. CONCLUSION

The current research is the empirical analysis of the generative model NB and various models such as LR, DT, RF, SVM, GB and ANN. Among the entire size of the KDD cup dataset, only 10 percent of the data size only taken into account for the empirical analysis. Among the models the RF ensemble model has presented the highest accuracy, the rest of the discriminative models also provide respectable results. As a future enhancement the entire dataset can be considered with the DL models and the Transformer-based pre-trained models can be taken into account for the execution and evaluation.

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